

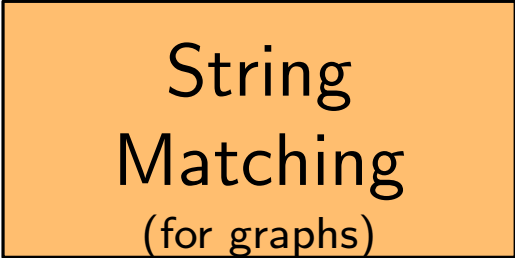
Distributed Quantum Computing

About Me

Three main research topics

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An orange rectangular box with a black border, containing the text 'String Matching (for graphs)'.

String
Matching
(for graphs)

About Me

Three main research topics

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Quantum
Computing

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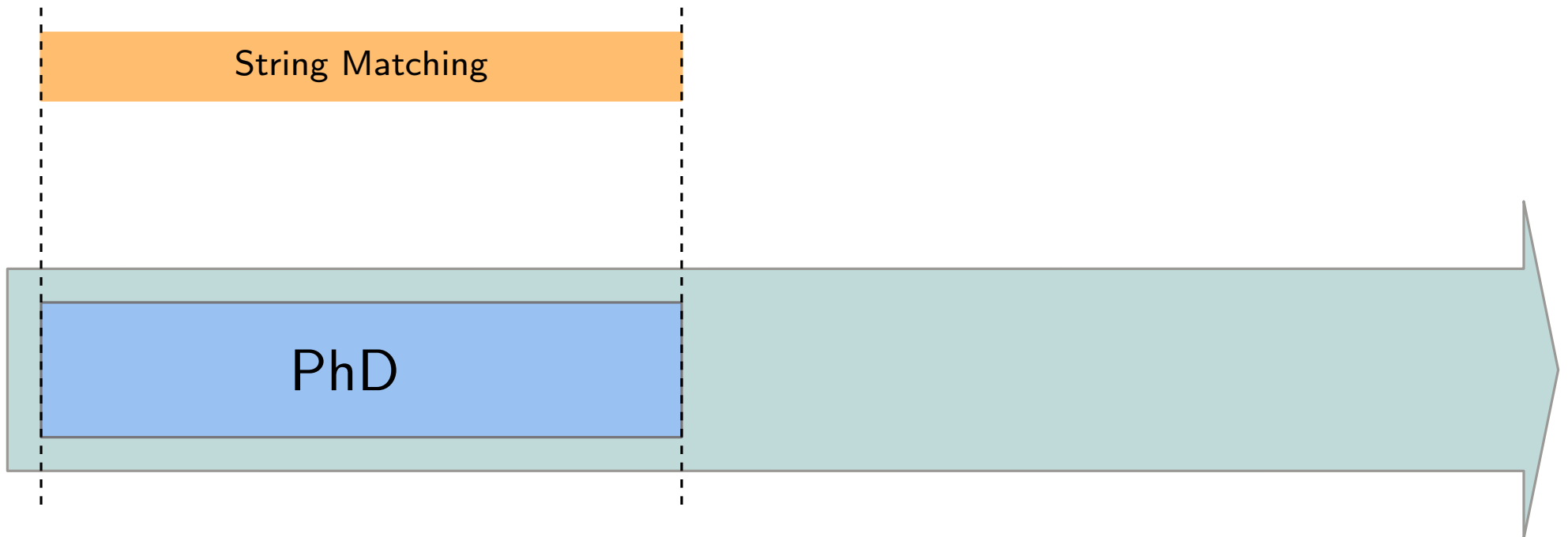
Distributed
Computing

String Matching

PhD

2019

2022



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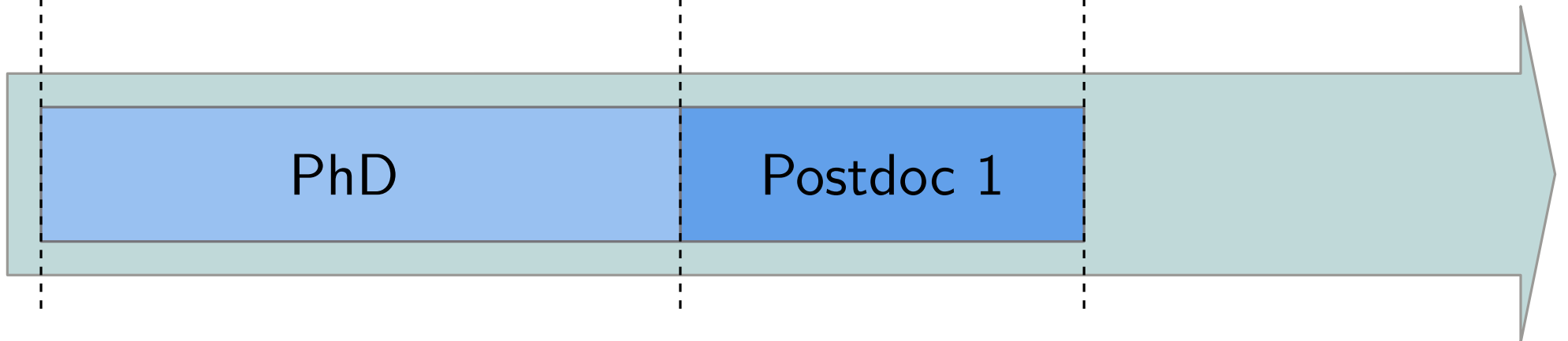
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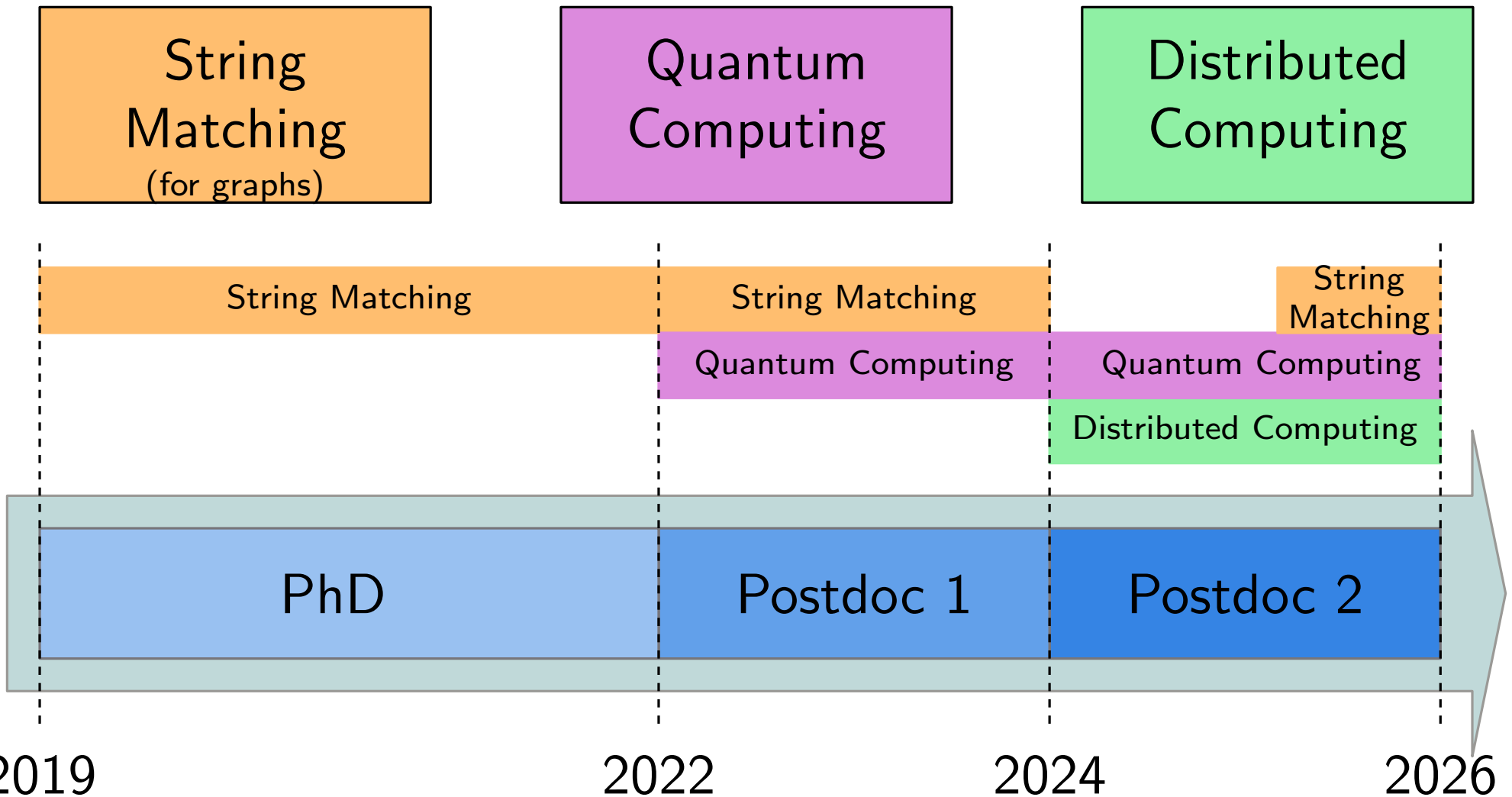
Postdoc 2

2019

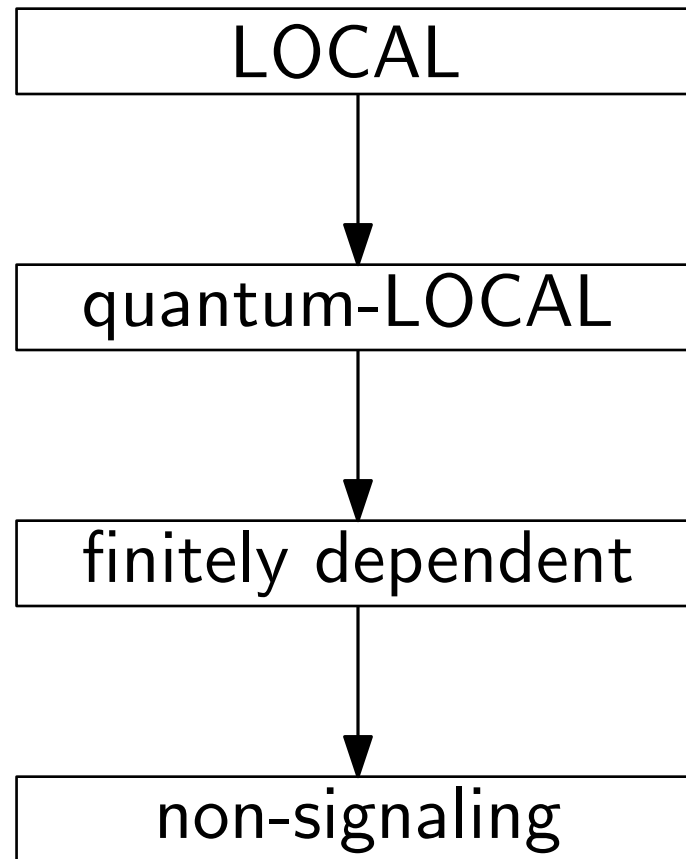
2022

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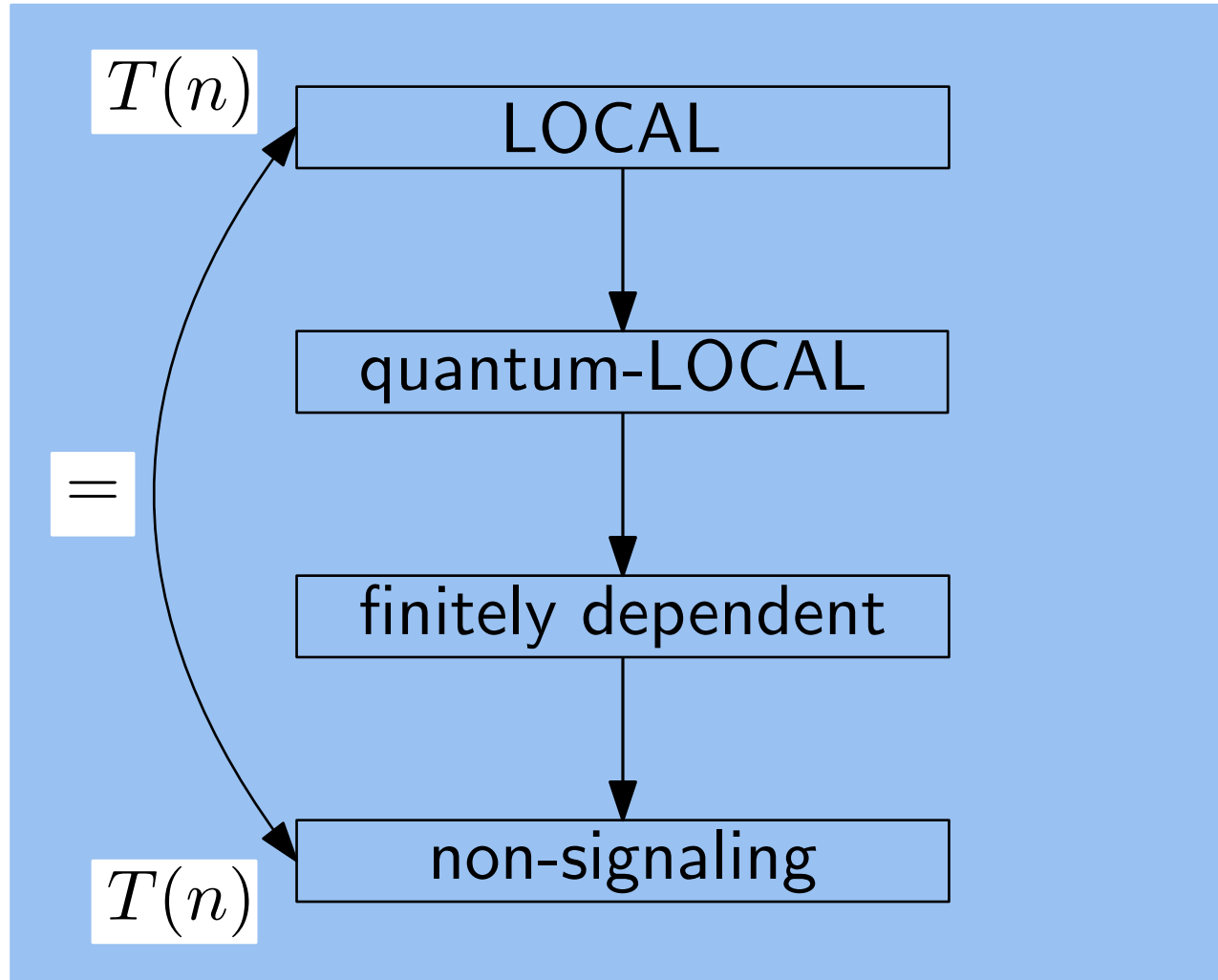


Models of Computations



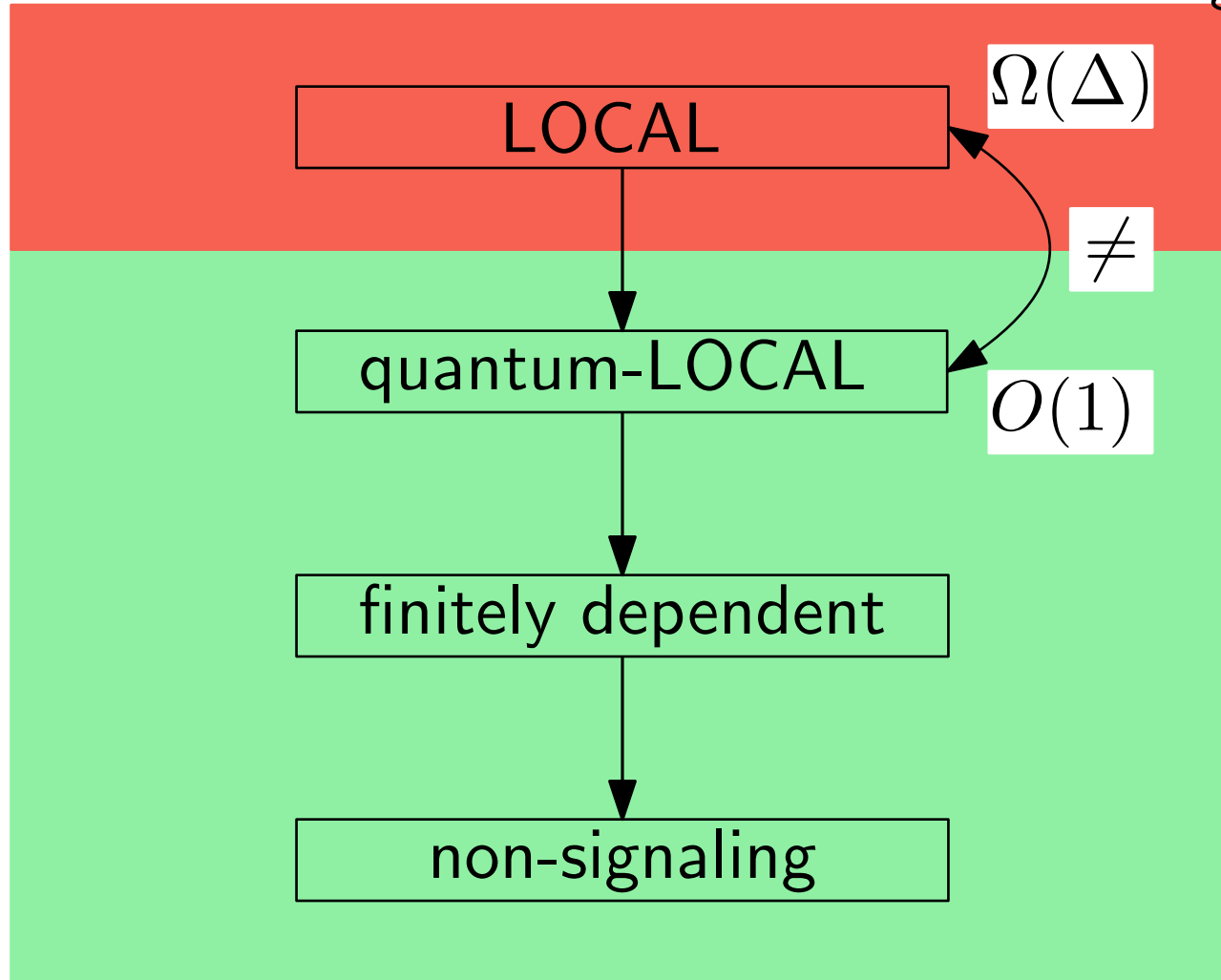
Models of Computations

Linear Programs (LPs)

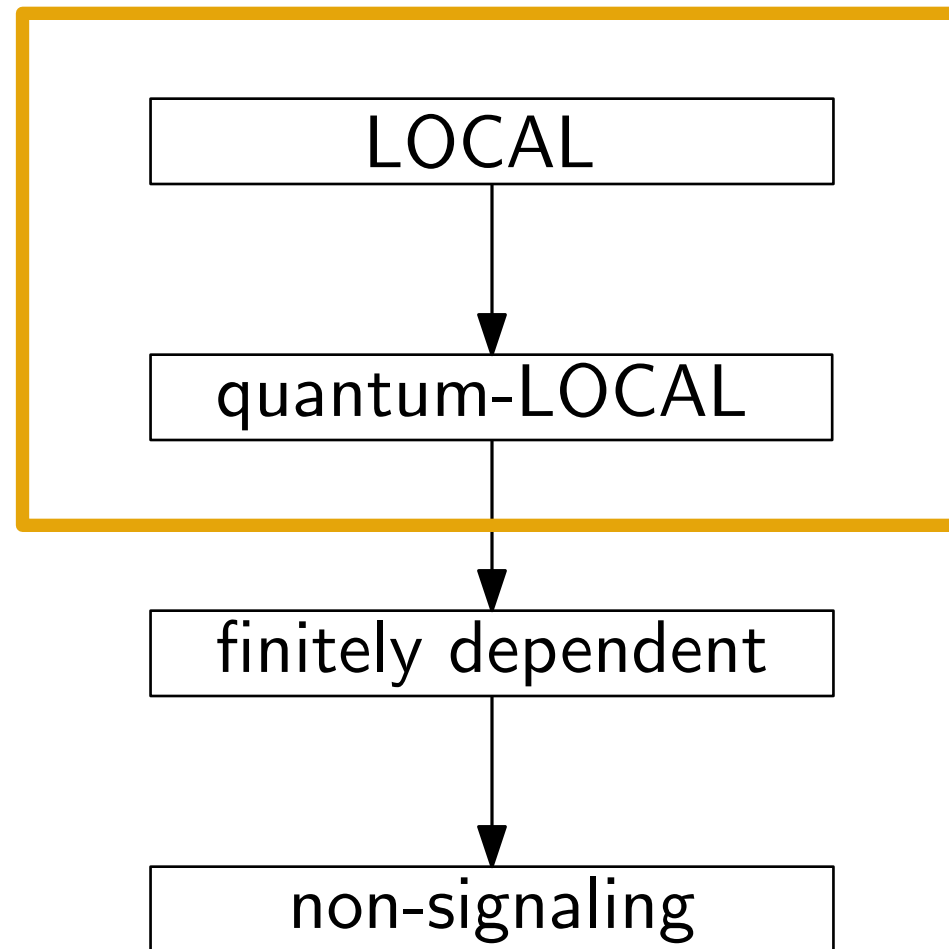


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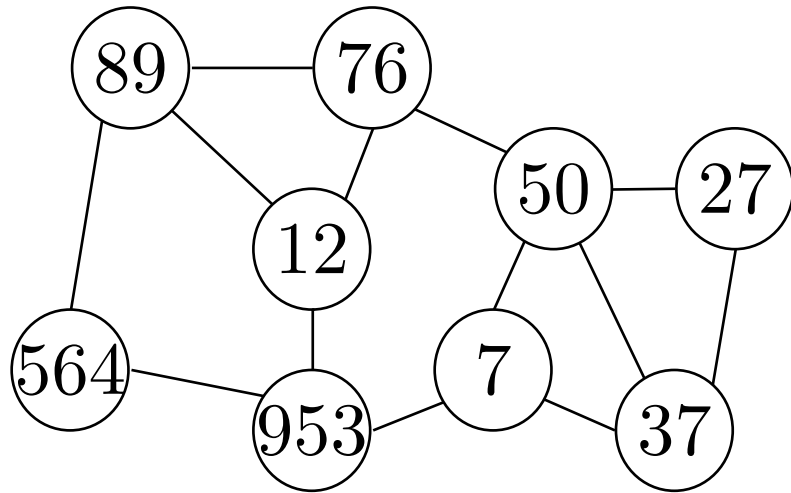
Locally Checkable
Labelling Problems



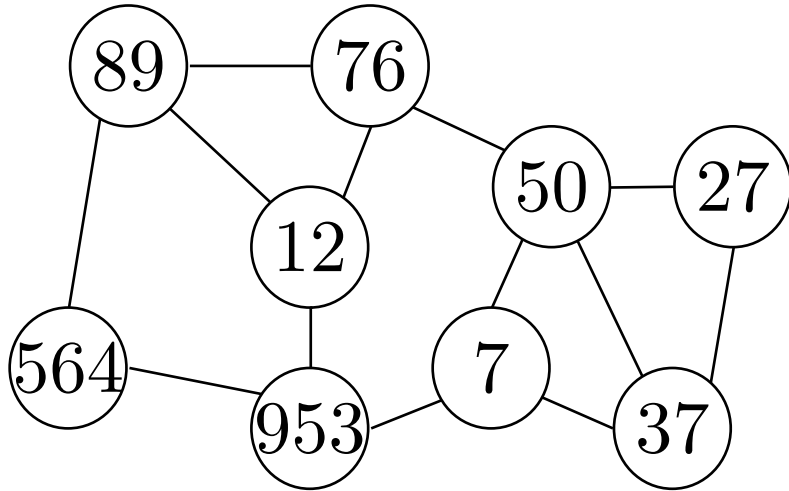
Models of Computations



LOCAL and quantum-LOCAL

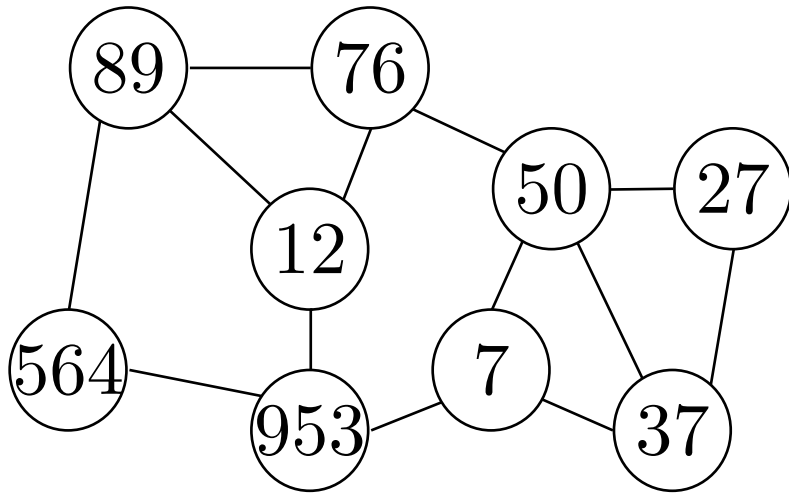


LOCAL and quantum-LOCAL



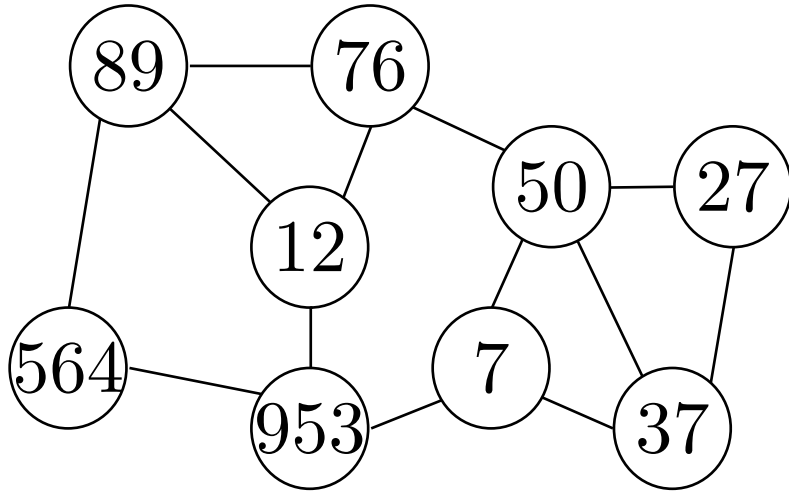
- The network is a graph $G = (V, E)$ of n nodes

LOCAL and quantum-LOCAL



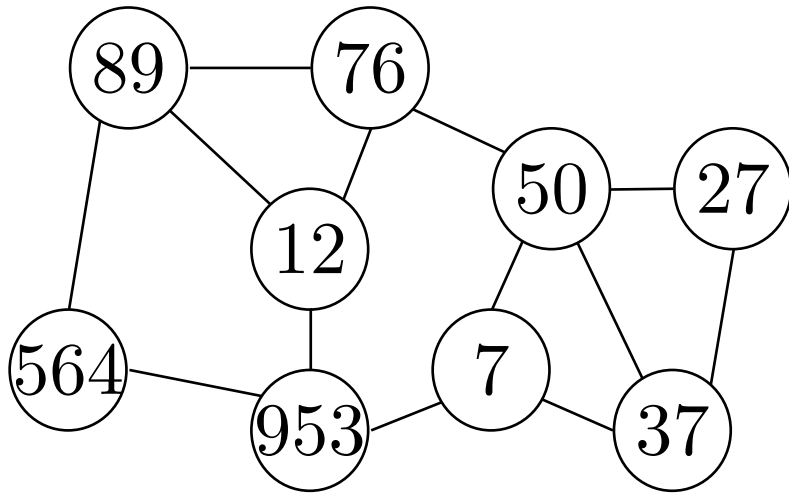
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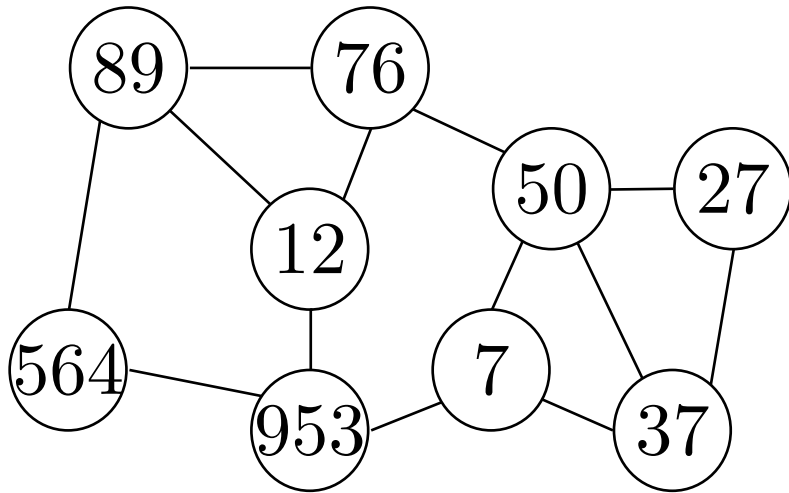
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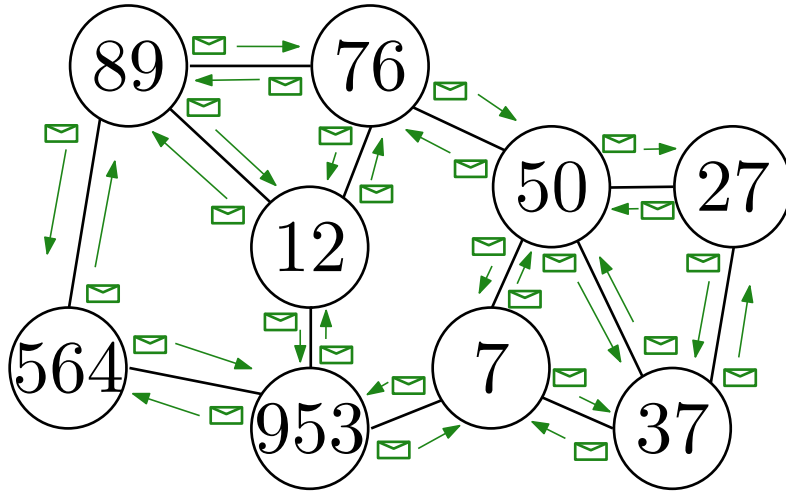


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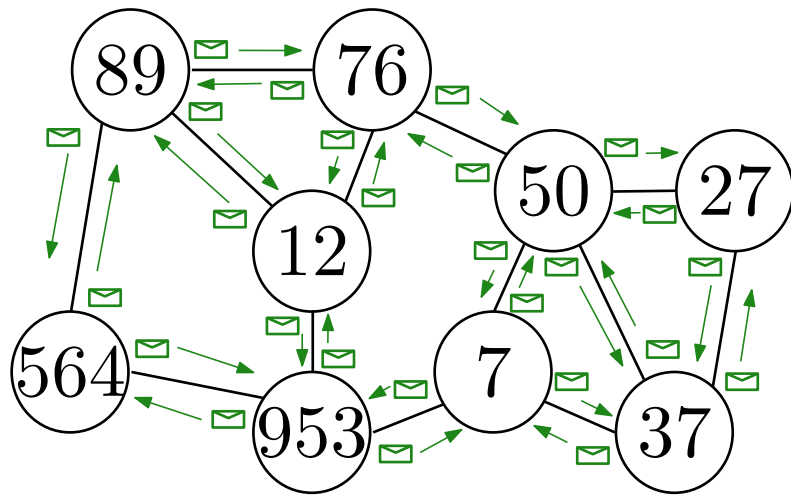
Complexity

Number of communication rounds as a function of n



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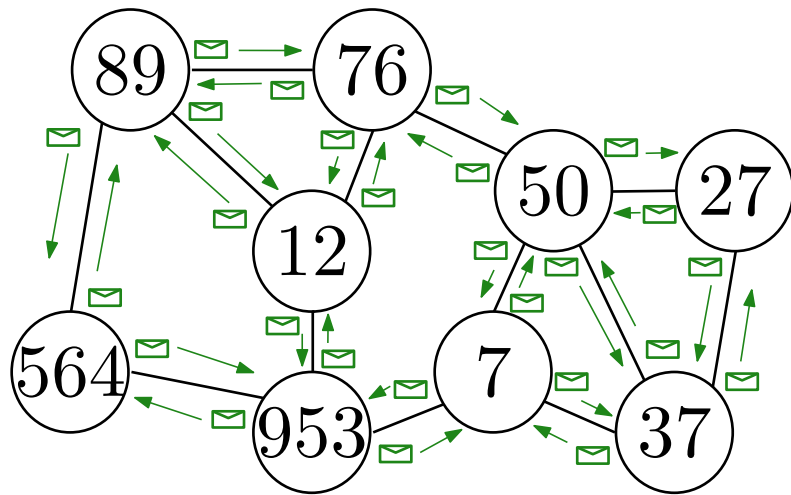
Number of communication rounds as a function of n

LOCAL

Nodes hold, send and receive **bits**

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LOCAL and quantum-LOCAL



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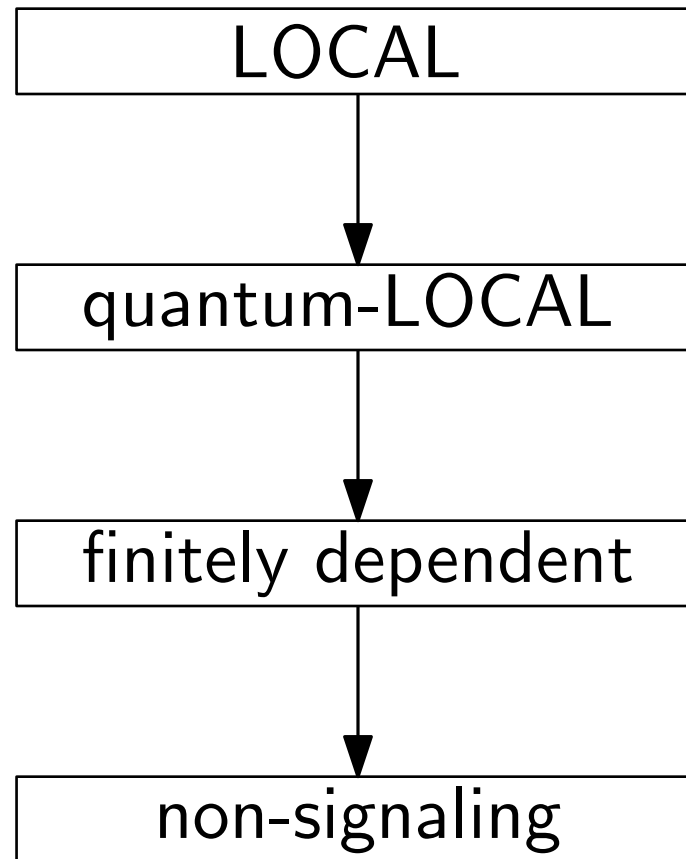
quantum-LOCAL

Nodes hold, send and receive **qubits**

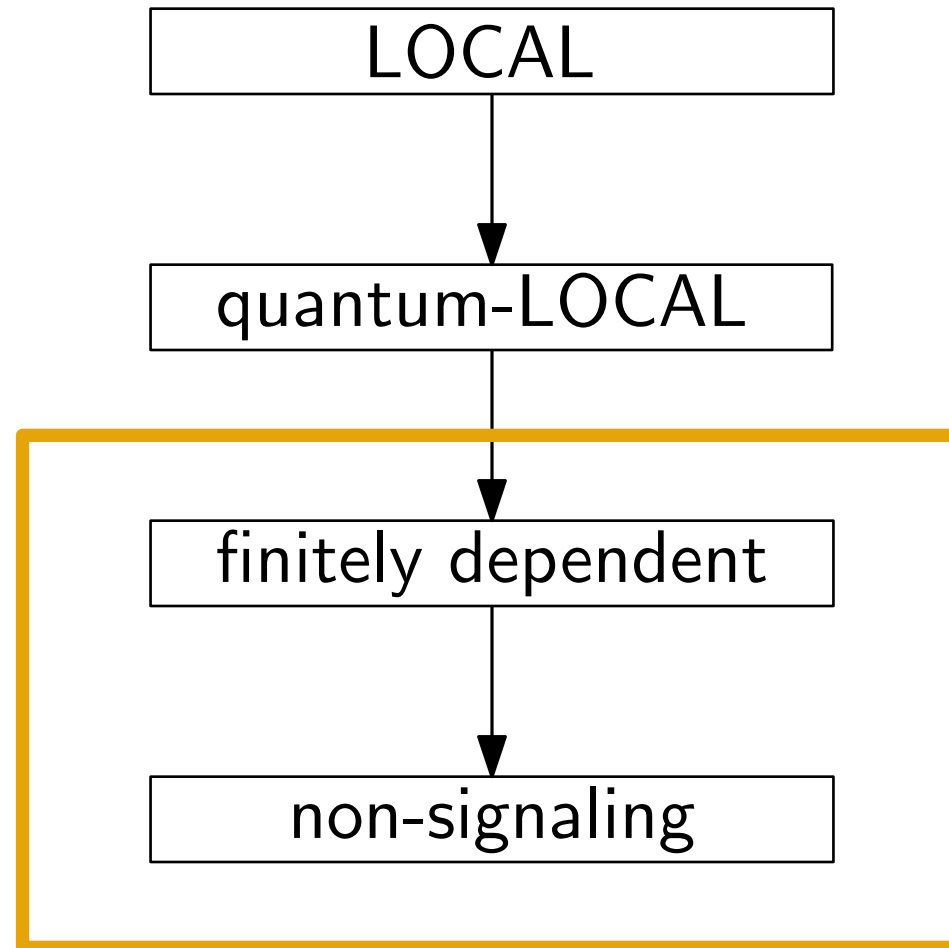
IDs are not needed

“send” qubits means applying SWAP gates

Models of Computations

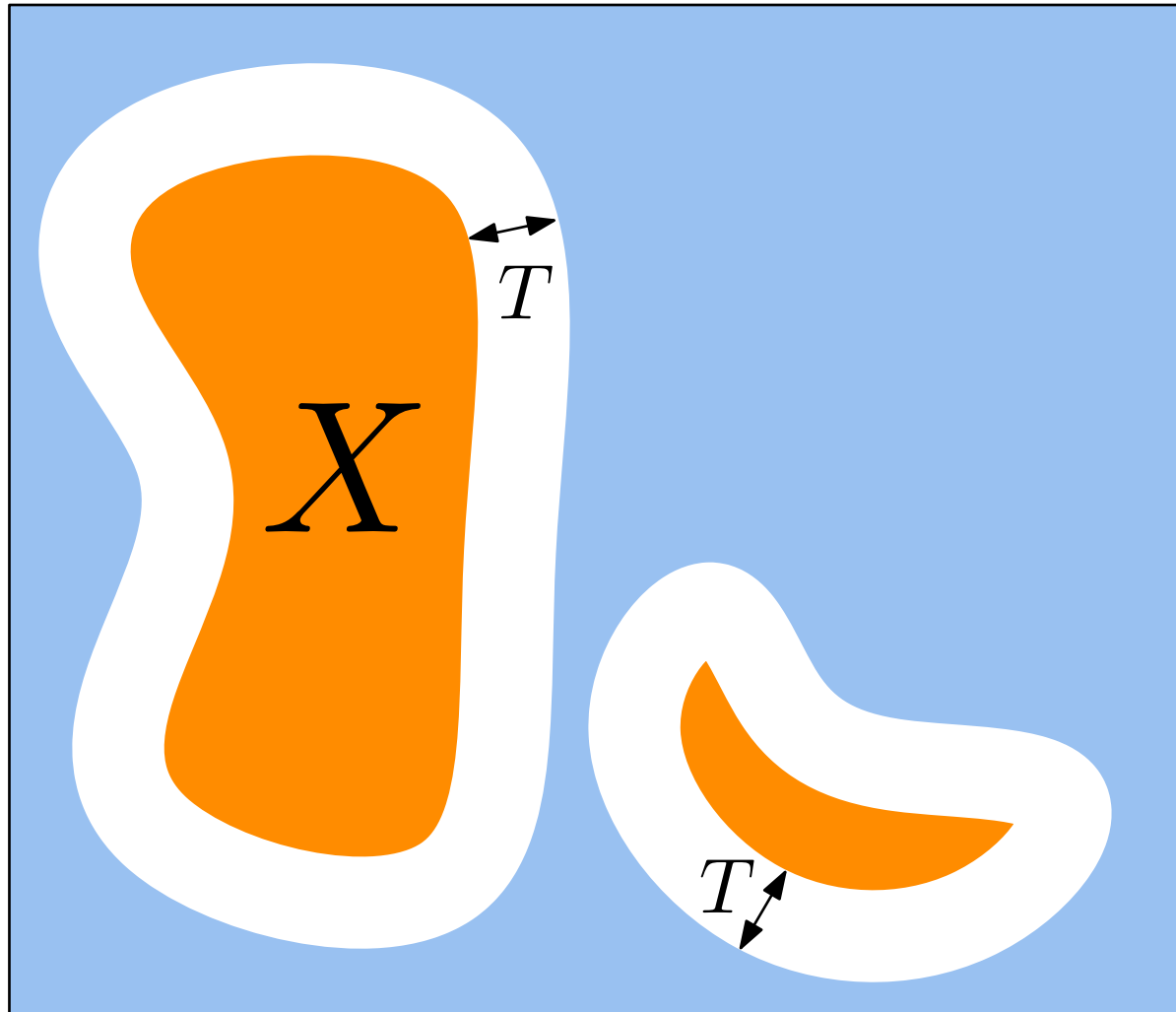


Models of Computations



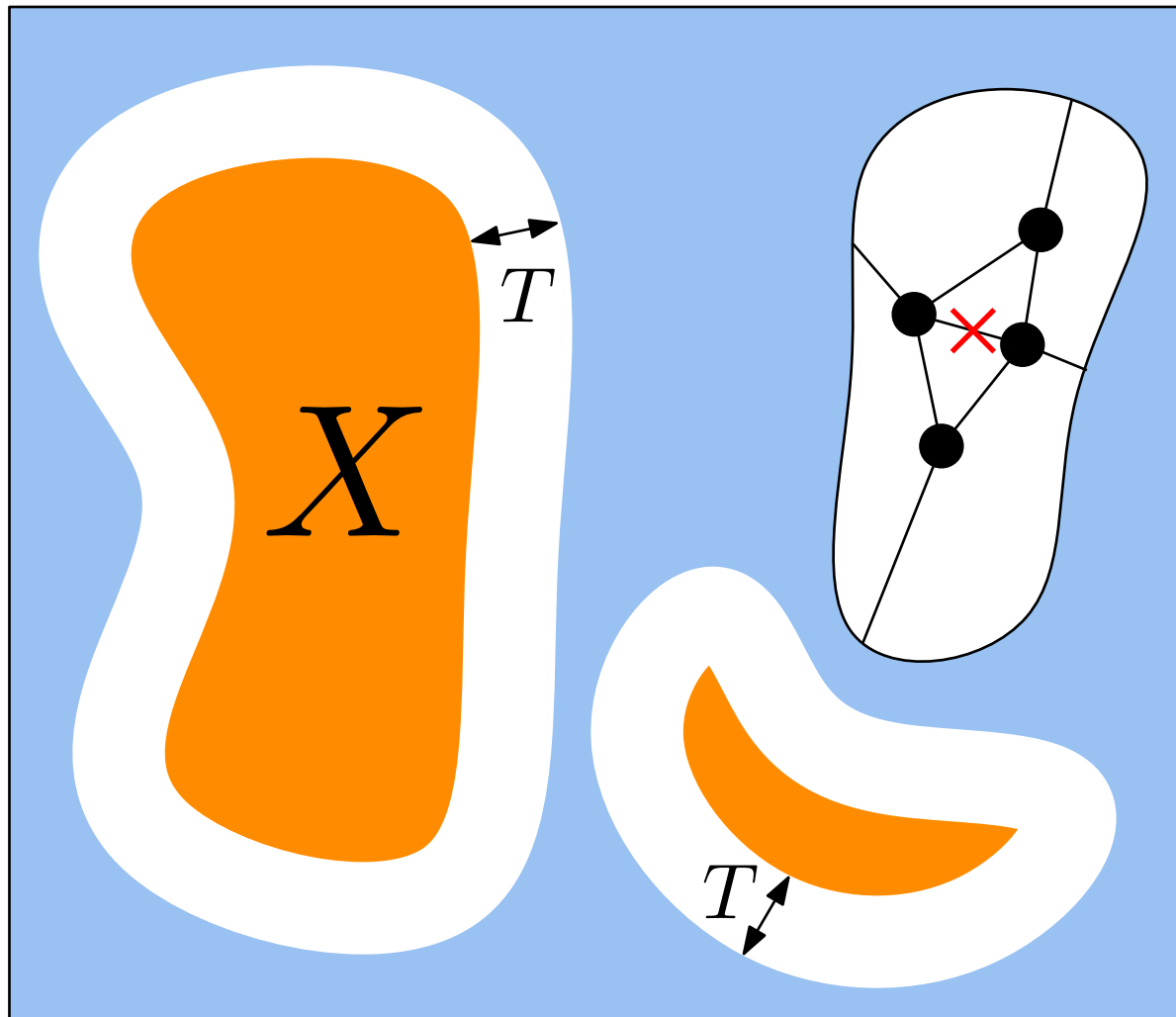
Non-signaling distributions

For any subset of nodes X , changes at distance $> T$
do not affect the output distribution of X



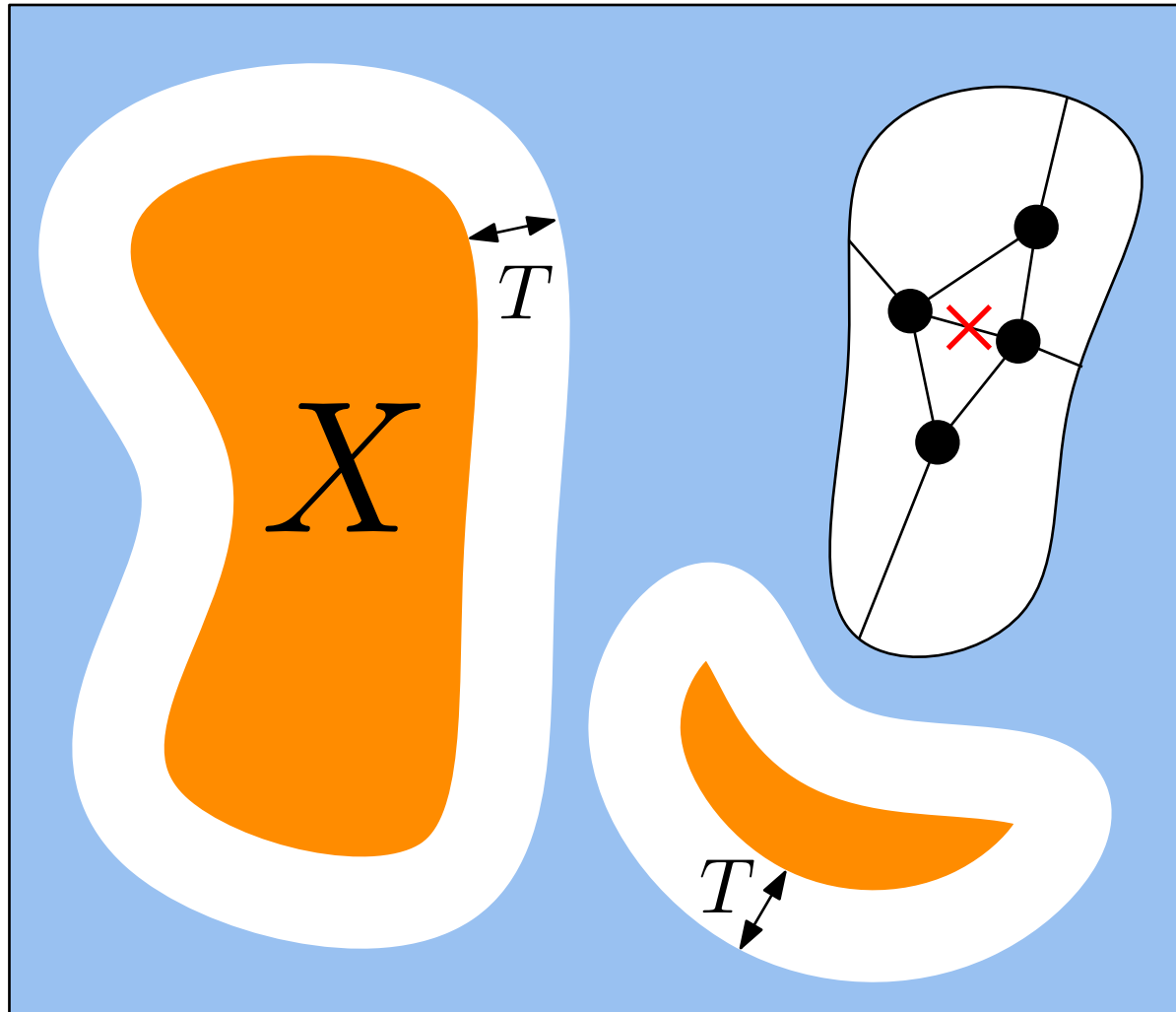
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Finetely dependent distributions add restrictions not considered today

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS):**

a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$

- **Maximal Independent Set (MIS):**

a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I, \exists v \in I : (v, u) \in E$

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Example of Distributed LP: **fractional maximum independent set**

$$\text{maximize } \sum_{v \in V} x_v$$

$$\text{subject to } \sum_{(v,u) \in E} x_v + x_u \leq 1 \quad \forall v \in V$$

$$0 \leq x_v \leq 1 \quad \forall v \in V$$

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When $x_v \in \{0, 1\} \quad \forall v \in V$, the solution is a **maximum independent set**, i.e. an MIS of maximum cardinality.

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v , explore your radius- $T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Here X_v is the rand. var. for the output of node v

Let $X = \sum_{v \in V} X_v$ be the rand. var. for the output of the entire graph

Key facts

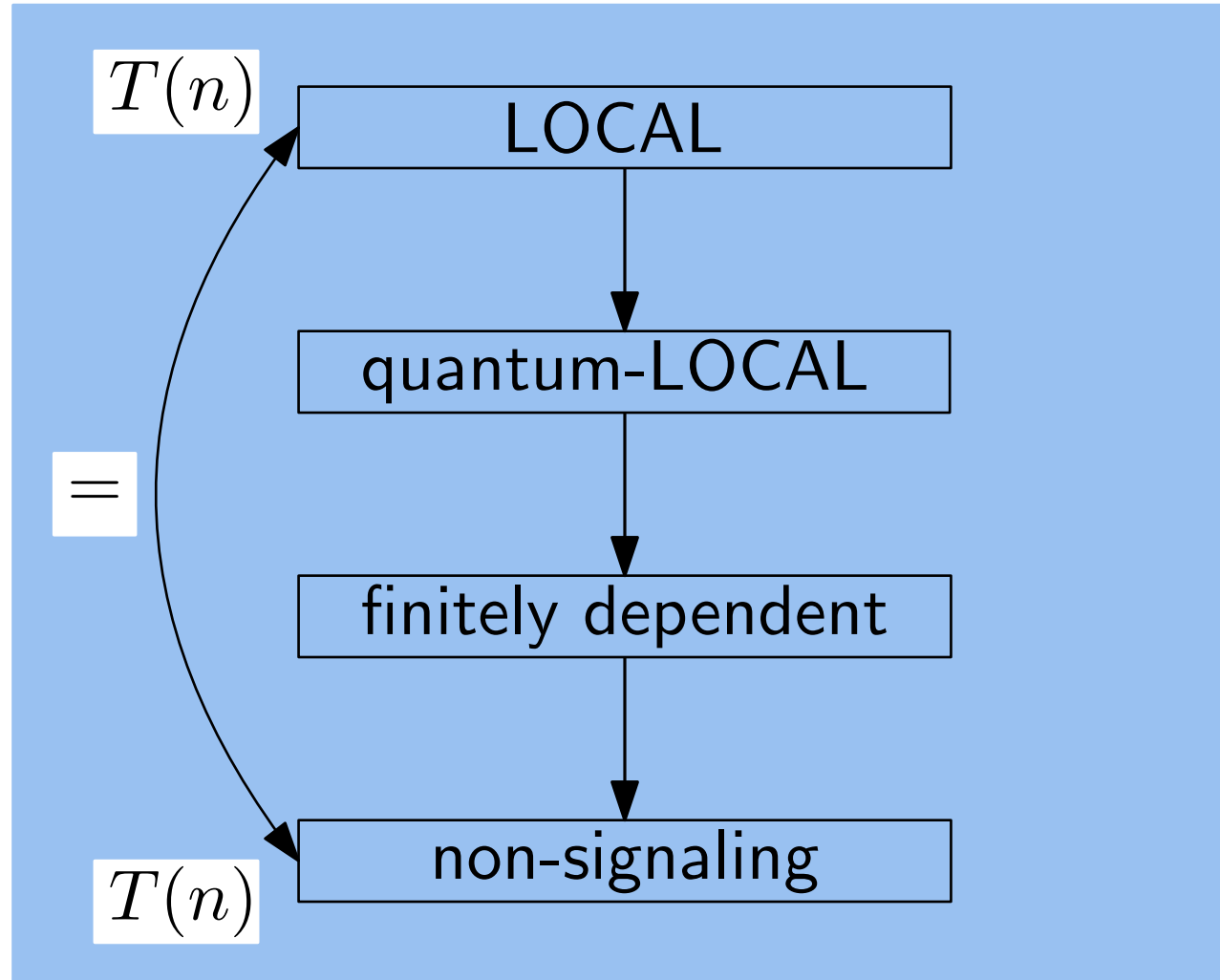
- By linearity of $\mathbb{E}$,

$$\sum_{v \in V} \mathbb{E}[X_v] = \mathbb{E} \left[\sum_{v \in V} X_v \right] = \mathbb{E}[X]$$

- By monotonicity of $\mathbb{E}$, if X is an α -approx. then $\mathbb{E}[X]$ is an α -approx.

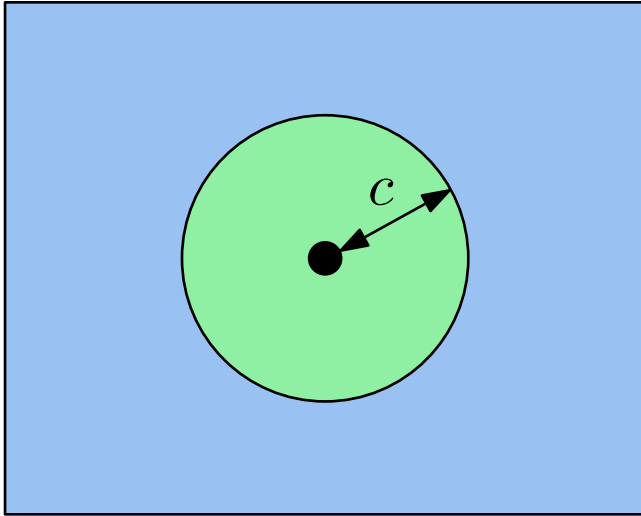
Results for Linear Programs (LPs)

Linear Programs (LPs)



Results for Locally Checable Labelly Problembs (LCLs)

LCLs: explore up to radius c to check a solution



Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

Reminder on GHZ

Three players A , B and C , and one referee

inputs x , y and z ; outputs: a , b and c

Goal: **without communicating after the input is give**, output bits such that

$$a \oplus b \oplus c = x \vee y \vee z$$

knowing that $x \vee y \vee z \in \{(0, 0, 0), (1, 1, 0), (1, 0, 1), (0, 1, 1)\}$